

# Africast-Tidy Time Series & Forecasting Using R

<https://workshop.f4sg.org/africast/>



# Project lead and Instructors



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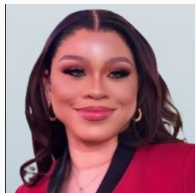


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# Mentors

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- Akounda Badibassa
- Mihertab Ermias
- Agnes Njambi
- Caroline Mugo
- Catherine Chepkoech Munga
- Charity Wamwea

# Mentors

- Dickson Ochichi Suguku
- Herbert Imbogah
- Matabel Odin
- Simon Kamau
- Susan Mwelu
- Abiodun Egwuenu
- Adamu Saidu Saleh
- Adebayo, Olatunji Joyce
- Anthony C. Uwandu-Uzoma

# Mentors

- Haruna Jimoh
- Hassan Adam Murtala
- Okorie, Oluchukwu Chiamaka
- Olagbami Oluwafemi Samson
- Ugonma Regina Okorie
- Shittu Ola Abdulganiy
- Ali Mohamed Ali
- Kundu Sam
- Ssembatya Ronald

# Coordination and admin lead

- Henry Kissinger Ochieng
- Caroline Mugo
- Winnie Chacha
- Samuel Mwalili

# Assumptions

- This is not an introduction to R. We assume you are broadly comfortable with R code, the RStudio environment and the tidyverse.
- This is not a statistics course. We assume you are familiar with concepts such as the mean, standard deviation, quantiles, regression, normal distribution, likelihood, etc.
- This is not a theory course. We are not going to derive anything. We will teach you time series and forecasting tools, when to use them, and how to use them most effectively.

## Key reference

**Hyndman, R. J. & Athanasopoulos, G. (2021)**  
***Forecasting: principles and practice*, 3rd ed.**



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**[OTexts.org/fpp3/](https://otexts.org/fpp3/)**

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**Hyndman, R. J. & Athanasopoulos, G. (2021)**  
***Forecasting: principles and practice*, 3rd ed.**

**[OTexts.org/fpp3/](https://otexts.org/fpp3/)**

- Free and online
- Data sets in associated R package
- R code for examples

## Reference - Recommended

**Kolassa S., Rostami-Tabar B., & Siemsen, E. (2023)**  
***<https://dfep.netlify.app/>, 1st ed.***

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**Kolassa S., Rostami-Tabar B., & Siemsen, E. (2023)**  
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## Reference - Recommended

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**<https://dfep.netlify.app/>**

- Free and online
- Not an in-depth technical book
- Mindset behind forecasting
- Overview of forecasting methods and processes

# International Institute of Forecasters (IIF)

- Certificate will be provided by IIF. You need to attend all lectures and exercise sessions to get the certificate.
- a nonprofit organization founded in 1982, is dedicated to developing and furthering the generation, distribution, and use of knowledge on forecasting

# Forecasting for Social Good committee

- Follow us on LinkedIn
- Follow us on Twitter
- Use #AFRICAST, #F4SG if you post on social media about the workshop

# Poll: How proficient are you in using R?

- 1 Guru: The R core team come to me for advice.
- 2 Expert: I have written several packages on CRAN.
- 3 Skilled: I use it regularly and it is an important part of my job.
- 4 Comfortable: I use it often and am comfortable with the tool.
- 5 User: I use it sometimes, but I am often searching around for the right function.
- 6 Learner: I have used it a few times.
- 7 Beginner: I've managed to download and install it.
- 8 Unknown: Why are you speaking like a pirate?



# Poll: How experienced are you in forecasting

- 1 Guru: I wrote the book, done it for decades, now I do the conference circuit.
- 2 Expert: It has been my full time job for more than a decade.
- 3 Skilled: I have been doing it for years.
- 4 Comfortable: I understand it and have done it.
- 5 Learner: I am still learning.
- 6 Beginner: I have heard of it and would like to learn more.
- 7 Unknown: What is forecasting? Is that what the weather people do?

# Install required packages

```
install.packages(c(  
  "tidyverse",  
  "fpp3",  
  "GGally",  
  "sugrrants",  
  "astsa"  
))
```

# Approximate outline

| Session | Topic                                         | Chapter |
|---------|-----------------------------------------------|---------|
| 1       | 1. Basics of time series and data structures  | 2       |
| 1       | 2. Time series patterns and basic graphics    | 2       |
| 2       | 3. Transforming / adjusting time series       | 3       |
| 2       | 4. Computing and visualizing features         | 4       |
| 3       | 5. Basic modeling / forecasting               | 1,3,5   |
| 3       | 6. Forecasting with regression                | 7,10    |
| 4       | 7. Exponential smoothing                      | 8       |
| 4       | 8. ARIMA models                               | 9       |
| 5       | 9. Basic training and test accuracy           | 5       |
| 5       | 10. Residual diagnostics and cross validation | 5       |

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# Schedule for each session (UTC+3)

- Start at 10:00 a.m.
- Lecture 1: 10:00 - 11:30
- Break: 11:30 - 11:45
- Lecture 2: 11:45 - 13:15
- Exercises: 14:00 - 17:00 (on your own)
- Mentorship: 17:00 - 18:00

# Access materials

- AFRICAST workshop website
- Mentorship sessions and Q/A
- Slack